Better Than Guessing

Class 18 Instructor Notes | 40 minutes of a 75-minute period | Four-page handout | Laptops, teams of 2–3

Placement and purpose

Follows the regression lecture. Students already have r, and the lecture has shown that slope and intercept both fall out of it. This activity answers the question that comes next: **was the line worth fitting?**

The one idea: R^2 measures how much the line beat a flat guess at the mean — and it can be high for a model that is flatly wrong. The residual plot is what catches that.

> **Structural note.** The order here is inverted from the source notebook. Students do the full six-step calculation once, on the **cricket** data, and meet the quadratic example afterwards as a twist rather than as a warm-up. Two reasons: it fits 40 minutes (one walkthrough, not two), and it makes the punchline land harder — the fake data set gets the **higher** R^2 (0.923 vs 0.698) while being the case where a straight line is plainly wrong.

Timing

6 min	Part 1 — two baselines, on paper, no code
20 min	Part 2 — SS_{Total}, fit, SS_{Residual}, $R^2 = r^2$
8 min	Part 3 — extrapolation and the negative R^2
6 min	Part 4 — the twist and the residual plots

> **If you are behind:** Part 3 is the one to compress — run the two cells from the front and assign Q3.1–3.2 as written reflection. Do not cut Part 4; the residual plot is the only diagnostic in the activity that generalizes beyond this data set.

Code blanks

Four, all in Part 2 except none elsewhere:

```
SS_total   = np.sum(deviations ** 2)
residuals  = temp - predicted
R_squared  = 1 - (SS_residual / SS_total)
```

(The second blank is the one students get wrong — they subtract the mean again instead of the prediction. The comment warns them; some will still do it.)

6 min — Part 1. Two Baselines

Paper only. **Q1.1:** the mean, because it minimizes total squared error — no other single number sits closer to all the points at once. **Q1.2:** perfect line $\longrightarrow SS_{\text{Residual}} = 0$, $R^2 = 1$; useless line $\longrightarrow SS_{\text{Residual}} = SS_{\text{Total}}$, $R^2 = 0$.

Getting Q1.2 down **before** any computation is what makes the negative R^2 in Part 3 register as strange rather than as just another number.

20 min — Part 2. How Good Is the Cricket Line?

quantity	value
mean temperature	80.04 °F
SS_{Total}	629.84
slope	3.2911

quantity	value
intercept	25.2323
SS_{Residual}	190.55
R^2	0.6975
r	0.8351

Q2.2 slope in plain English: each additional chirp per second corresponds to about **3.3 °F** warmer. Push for the units — "degrees Fahrenheit per chirp per second" is the answer, and students who cannot state them usually cannot interpret the number either.

Q2.3: chirp rate accounts for about **70%** of the variation in temperature; the remaining 30% is variation the line cannot explain — measurement error, other factors, and the fact that crickets are not thermometers.

Q2.4 is the calibration question and worth the time. Last class students found $r = 0.802$ between Pixar rating services and called it strong. Squared, that is **0.64** — barely two-thirds of the variation. Most students carry an intuition that $r = 0.8$ means "80% of the way to perfect." It does not. The crickets make the same point: $r = 0.84$ sounds excellent, $R^2 = 0.70$ sounds noticeably less so, and they are the same fit.

8 min — Part 3. Where the Line Stops Working

Predictions: **19 chirps → 87.8 °F** (fine, inside the observed range of 14.4–20.0); **40 chirps → 156.9 °F** (absurd — and no cricket chirps that fast); **0 chirps → 25.2 °F**, the intercept itself.

The intercept is the better half of this question. Extrapolating to 40 is an obvious mistake that no one would make by accident. But the intercept is **printed in the regression output** and looks like a legitimate result — and it claims something about a temperature at which crickets have been silent for thirty degrees. Students who write "the intercept is the temperature when there are no chirps" have stated the arithmetic correctly and the science wrongly.

Q3.2: forcing the line through the origin while keeping the fitted slope makes it predict **49.4 °F at 15 chirps/sec**, against actual temperatures of 69–93 °F. It is 20–30 degrees low across the whole data set, so SS_{Residual} balloons to 9741 against 190 for the real line, and R^2 falls to about **−14.5**. Negative R^2 means the flat line at the mean would have been better.

Expect the sharp question: "why not refit the slope through the origin?" The honest answer is that you can, and it gives slope 4.79 and $R^2 = +0.55$ — positive. So this demonstration is **not** showing what a through-origin regression does; it is showing what happens when you keep a slope that was fitted alongside an intercept and then throw the intercept away. The general lesson survives intact: R^2 compares whatever prediction rule you hand it against the mean, and a bad rule loses. Do not claim more than that.

6 min — Part 4. A Better Number That Is a Worse Model

Second data set: $x = 0..6$, $y = x^2$. Fitted line $y = 6x - 5$, $SS_{\text{Total}} = 1092$, $SS_{\text{Residual}} = 84$, $\boldsymbol{R^2} = \mathbf{0.9231}$.

Q4.1: the quadratic data gets the higher R^2 (0.923 vs 0.698), and it is the one where a straight line is the **wrong** model. The crickets have a genuinely linear relationship with real scatter; the second set has no scatter at all and a systematic curve.

Q4.2 — the residuals. This is the payoff.

x	0	1	2	3	4	5	6
residual	+5	0	−3	−4	−3	0	+5

A perfect U. The cricket residuals, by contrast, scatter from −6.5 to +5.0 with no pattern — their correlation with chirp rate is exactly 0, which is true of **any** least-squares fit by construction, so what matters is the visible shape, not the number.

Q4.3: no. A high R^2 says the line is closer to the points than a flat mean is; it says nothing about whether a line was the right shape. The residual plot shows structure the line failed to capture — and here it shows it loudly, on the data set with the better R^2.

Closing line worth saying out loud: every regression you fit from here on gets a residual plot before you report the R^2. It costs one cell.

Notes on what was left out

Bootstrapping. Cricket_Thermometer.ipynb continues into bootstrap confidence intervals for the slope and a seaborn regplot band. That needs regression to be settled first, so it belongs in a later class. Its Challenges 1–3 — explain intercept(), predict at 19 chirps, why not extrapolate to 40 — are folded into Parts 2 and 3 here.

The Cars 2 callback. Class 17's notes promised that the diagnostic catching Cars 2 would arrive this class. Residual plots deliver only half of that: on the Pixar fit, Onward has the larger residual (−18.0) and Cars 2 the larger **leverage** (0.599 against a mean of 0.095). If you want to close that loop, it takes a fifth part introducing leverage as "how far this point sits from the center of x," which does not fit 40 minutes. Otherwise say plainly that the second half of the answer is still coming.